Real-Time Anxiety Recognition through Optical-Flow Based Apex-Phase Spotting and Deep Learning

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Abstract: Real-time micro-expression analysis remains challenging due to the subtle intensity and extremely short duration of facial movements, making accurate spotting and recognition difficult. Recent studies have explored optical flow, Region of Interest (ROI)-based analysis, and deep learning for micro-expression recognition. However, most existing approaches primarily focus on offline emotion recognition and rarely integrate automatic apex spotting with real-time anxiety recognition in a unified and computationally efficient pipeline. This study proposes an end-to-end web-based framework for real-time anxiety recognition using facial micro-expressions. The proposed framework combines WebRTC-based video acquisition, ROI-constrained TV-L1 optical flow for motion representation, automatic apex spotting using prominence-based peak detection, and temporal deep learning models for anxiety classification. Four architectures, namely Temporal CNN, BiLSTM, Temporal Convolutional Network (TCN), and Transformer, were evaluated. Experimental results show that the proposed apex spotting method achieved Mean Absolute Error (MAE) values of 12.74 and 5.58 frames on the CASME II and SAMM datasets, respectively. Among the evaluated models, CNN Temporal achieved the highest validation accuracy of 80.00%, while the Transformer demonstrated the best cross-domain generalization with 66.67% accuracy on an independent test set. Furthermore, the complete pipeline achieved an end-to-end latency of 65.67 ms, demonstrating its feasibility for real-time anxiety recognition through a web-based platform.

Keywords: Micro-expression Recognition, Anxiety Recognition, Optical Flow, Apex Phase Spotting, Real-Time Processing

1 Introduction

Anxiety is an internal emotional state that is difficult to observe reliably through casual observation, largely because individuals can consciously suppress or mask outward expressions of how they feel. Yet the true emotional state often leaks through in brief, subtle facial changes known as emotional leakage [1]. This phenomenon suggests that micro expressions can serve as a behaviorally relevant indicator for uncovering latent emotional states that are otherwise difficult to conceal deliberately [2]. The relevance of this line of research has grown considerably across evaluative settings such as job interviews, presentations, and question and answer sessions. In the context of the transition into professional life, anxiety commonly arises from uncertainty about one's career future [3], [4]. Situational pressure during communication has likewise been shown to trigger anxiety responses that influence an individual's behavior throughout an interaction [5].

By definition, micro expressions are extremely brief, typically lasting between 0.04 and 0.3 seconds and are characterized by low-intensity, spatially localized facial movement [6]. These characteristics make micro expressions difficult to observe consistently, even for trained human observers. Consequently, a substantial body of research has turned to computer based approaches for detecting facial changes objectively and automatically. Among the available techniques, optical-flow-based motion representation has emerged as one of the most widely adopted approaches, owing to its ability to effectively capture both the direction and magnitude of facial movement between frames [7]. In addition, constraining analysis to active Regions of Interest (ROI) such as the eyebrows, eyes, and mouth has been shown to simultaneously increase detection sensitivity and reduce computational complexity [8], [9].

As micro-expression research has matured, a range of approaches based on optical flow, expression-phase detection, and deep learning have been developed to capture subtle facial motion and automatically recognize expression patterns [10], [11], [12]. However, the majority of existing work remains confined to recognizing basic emotions under controlled laboratory conditions and is processed offline. What remains conspicuously underexplored is the use of micro-expressions to validate anxiety states under genuinely real-time conditions and, more specifically, the integration of apex spotting, motion representation, and temporal classification into a single, efficient, end-to-end pipeline capable of operating live. This gap is precisely where the present work is positioned—not merely as an incremental improvement to any one component, but as a demonstration that these components can be unified into a coherent real-time system without sacrificing either detection accuracy or responsiveness.

To address this gap, we propose an anxiety recognition framework built on facial micro expressions that combines TV-L1 optical-flow motion representation, spatial constraint via facial ROIs, prominence based micro expression phase detection with a dynamic threshold, and classification through a purpose-built Temporal CNN. TV-L1 optical flow was selected for its capacity to maintain stable motion estimation under low-amplitude facial changes and its robustness to noise, illumination variation, and outlier data [13], [14]—properties that are essential when the signal of interest is, by definition, faint and fleeting.

The main contributions of this study are as follows: (1) an efficient, ROI-based motion representation designed explicitly for real-time micro-expression analysis, (2) a fully automatic apex-phase spotting method, evaluated on the CASME II and SAMM benchmark datasets, that removes the need for manual annotation at inference time, (3) a comparative evaluation of several temporal deep learning architectures for anxiety classification, including a cross domain generalization test that is rarely reported in prior real-time micro expression work, and (4) the implementation and empirical evaluation of a complete end-to-end pipeline that recognizes anxiety in real time. Taken together, these contributions position the proposed system as one of the few reported pipelines that jointly achieve automatic apex spotting, anxiety specific classification, and sub 100 millisecond end-to-end latency within a single deployable architecture, the core novelty this paper puts forward for real-time micro-expression recognition.

2 Related Works

Micro expression detection remains a substantially difficult task, and reliably identifying the onset, apex, and offset phases is a prerequisite for any downstream analysis. A number of studies have developed spotting methods that exploit facial motion information to localize the onset of an expression change. Optical flow based approaches are commonly used because they can represent subtle motion that unfolds over an extremely short duration [11], [12]. Beyond improving the accuracy of phase detection, recent work has also placed growing emphasis on computational efficiency to support real-time deployment. In this context, methods based on apex frames, feature tracking, and phase representation have demonstrated the ability to reduce computational complexity without materially sacrificing detection performance [15], [16].

Recent studies have likewise increasingly relied on deep learning architectures capable of extracting information from micro-expressions effectively. Lightweight CNN architectures are widely used to strike a balance between accuracy and computational efficiency for real-time applications [17], [18]. Other work leverages attention mechanisms, hierarchical representations, and spatio-temporal feature-fusion strategies to improve a model's ability to recognize micro expressions from subtle motion representations [19], [20]. More recent trends also point toward Transformer based architectures and the integration of large language models to capture more complex temporal relationships and to improve the interpretation of an individual's emotional state [21], [22]. Notably, few of these efforts have been validated end-to-end under live, real-time operating conditions, a gap the present study directly targets.

3 Methodology

3.1 Overview

This study develops a facial-micro-expression based anxiety-recognition system that operates in real time through a web platform. The system pipeline consists of five main stages: facial video acquisition, motion extraction via TV-L1 optical flow, micro-expression phase detection, feature representation construction, and classification using a deep-learning model. The entire process runs end-to-end on the server, so that analysis results can be returned to the user immediately.

3.2 Data Acquisition and Preprocessing

Facial video is captured from the user's webcam through the web client. Each video clip is processed by detecting the face and extracting facial landmarks. The resulting landmarks are also used to perform face alignment, minimizing head pose variation so that the analysis remains focused on expression changes rather than head movement. Following alignment, several Regions of Interest (ROI) that play a central role in micro expression formation are selected: the eyebrow, eye, and lip regions.

Inter frame facial motion is represented using TV-L1 optical flow. This method was chosen because it produces stable motion estimates under low-amplitude facial changes and is more robust to noise and illumination variation than conventional optical-flow approaches—a property that is decisive when the target signal is a brief, low-amplitude micro-expression rather than an exaggerated macro-expression [13], [14].

3.3 Apex-Phase Spotting

Micro-expression detection is performed by analyzing the optical flow magnitude signal computed across all ROIs. Motion magnitude is accumulated into a temporal signal that describes the intensity of facial change throughout the video. Candidate apex frames are identified using a prominence based peak detection method, so that only genuinely significant motion peaks are retained.

Once the apex frame has been identified, the onset and offset phases are determined using a dynamic threshold computed from the local characteristics of the signal. This design allows the system to generate a complete onset-apex-offset phase representation automatically, without requiring manual annotation, an essential property for any system intended to operate in real time rather than in an offline, annotation-assisted setting.

3.4 Pipeline and Evaluation

The frame range obtained from the apex spotting process is used to form a single micro expression episode. From each ROI, a set of optical flow derived behavioral descriptors is extracted, as summarized in Table 1.

Feature Matrix	Description
Mean dx and dy	Average horizontal and vertical displacement of a facial region
Magnitude and Motion Energy	Intensity of facial movement
Direction Consistency	Consistency of facial motion intensity
Acceleration and Jerk	Temporal dynamics of motion change
Pairwise Synchronization	Degree of synchronization between facial regions
Facial Symmetry	Balance between the left and right sides of the face

Table 1: Description of Behavioral Feature Representation

The combination of features produces a representation that captures not only the magnitude of facial movement, but also inter-region coordination, temporal dynamics, and the balance of activity that emerges during an expression.

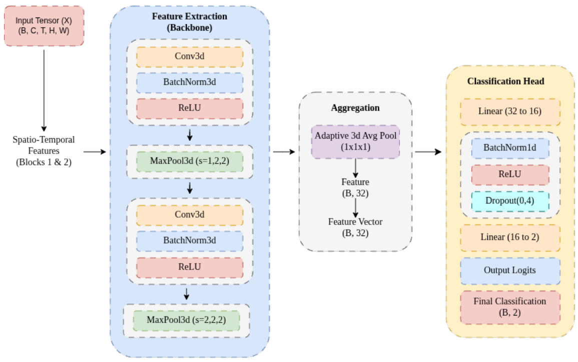


Fig. 1: Temporal CNN-based classification architecture.

The classification stage uses a CNN model designed to capture spatial and temporal patterns simultaneously. The model architecture is illustrated in Fig. 1.

In addition to the primary model, this study also conducts a comparative evaluation against several temporal models —BiLSTM, Temporal Convolutional Network (TCN), and Transformer—to analyze how different temporal-modeling strategies affect classification performance.

We also evaluate along two principal axes: micro expression phase detection and anxiety state classification. Apex spotting performance is evaluated using Mean Absolute Error (MAE) between the predicted and ground-truth apex frames on the CASME II and SAMM datasets. Classification performance, in turn, is evaluated using accuracy, precision, recall, F1-score, and balanced accuracy. Beyond accuracy metrics, the system is also tested under real-time conditions by measuring end-to-end latency, confirming that the entire pipeline can run continuously in a live production environment rather than only in a controlled, offline benchmark.

4 Experiments and Results

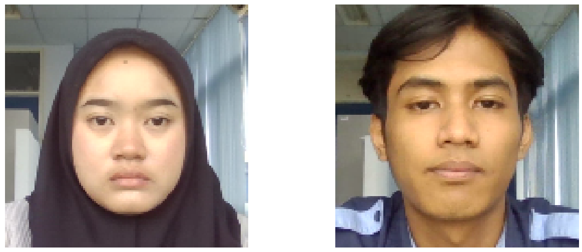
4.1 Apex-Phase Spotting

Experiments were designed to evaluate the system along three principal axes: micro expression phase detection capability, anxiety classification accuracy, and real-time processing performance. Phase-detection evaluation was carried out using the CASME II and SAMM benchmark datasets, which provide apex frame annotations as ground truth. Classification evaluation, meanwhile, used a primary dataset collected through facial-video acquisition under both anxious and non-anxious conditions.

Fig. 2 shows subject data for anxious and non-anxious conditions. Facial motion in this data was analyzed with attention restricted to the regions most relevant to micro expressions: the eyebrows, eyes, and lips, determined from facial landmark positions that produce a consistent ROI across every frame. The resulting selection of the most salient regions is shown in Fig. 3.

The extracted optical-flow field is shown in Fig. 4, computed using the TV-L1 algorithm, where each pixel represents the estimated magnitude of motion between two consecutive frames.

Micro expression phase analysis was then performed by accumulating optical flow magnitude values into a temporal signal, which was used to automatically identify the onset, apex, and offset phases via prominence based peak detection. The resulting phase detection output is shown in Fig. 5.



a) Anxious b) Non-anxious

Fig. 2: Illustration of anxious and non-anxious conditions.

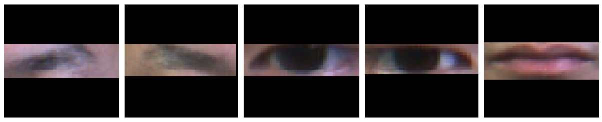


Fig. 3: Facial Regions of Interest: eyebrows, eyes, lips.

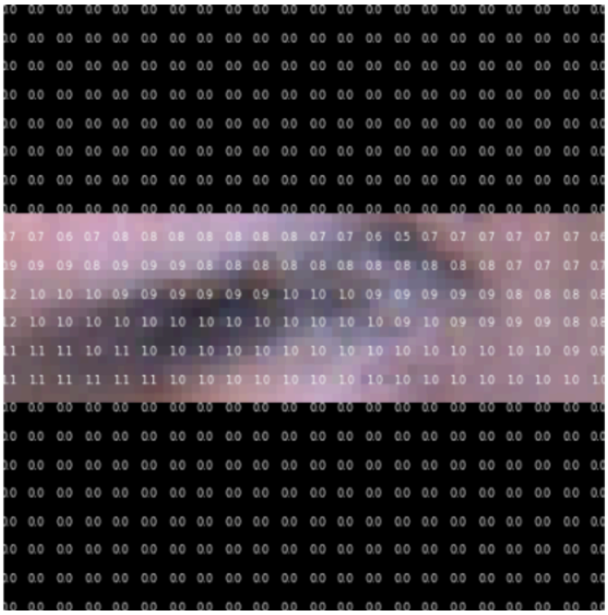


Fig. 4: Extracted Optical Flow.

Dataset	Sample	MAE (Frames)
CASME II	256	12.74
SAMM	159	5.58

Table 2: Spotting Result with Public Micro-Expression Datasets

Fig. 5: Micro Expression Phase Detection Result.

The evaluation results show that the apex spotting method achieves an MAE of 12.74 frames on CASME II and 5.58 frames on SAMM. These results indicate that the optical-flow magnitude signal is sufficiently reliable for automatically identifying the apex position with a relatively low error rate, a level of accuracy that is difficult to obtain without the ROI constrained, prominence based design proposed in this work.

4.2 Anxiety Classification

The output of micro expression phase detection was used to construct a facial behavior representation comprising mean displacement, magnitude, motion energy, direction consistency, acceleration, jerk, pairwise synchronization, and facial symmetry. This representation served as the input to the deep-learning models for classifying anxious versus non-anxious states.

Model	Data	Accuracy	Macro-F1
Temporal CNN	Validation	80.00%	0.7980
	Test	48.04%	0.4655
Bi-LSTM	Validation	65.00%	0.6419
	Test	73.68%	0.4815
TCN	Validation	60.00%	0.6703
	Test	64.71%	0.5354
Transformer	Validation	75.00%	0.7442
	Test	66.67%	0.5816

Table 3: Micro Expression Phase Detection Result

Temporal CNN attains the highest validation accuracy at 80.00% with a Macro-F1 of 0.7980, demonstrating its ability to learn micro expression patterns associated with anxiety. Its performance decreases on the independent test set, however, indicating a generalization limitation typical of CNN based temporal models trained on comparatively small, subject specific datasets.

The Transformer, in contrast, exhibits more stable performance under cross domain testing, achieving 66.67% accuracy on the test data. This suggests that attention based architectures such as the Transformer are better able to adapt to variation in subject-specific characteristics within temporal data compared with the Temporal CNN, an important finding for any system intended for real world, subject independent deployment rather than single-subject calibration.

4.3 Real-Time Performance

Stage	Latency
Landmark & ROI Detection	± 22.23 ms
Optical Flow TV-L1 Extraction	± 15.17 ms
Apex Phase Spotting	± 1.11 ms
Inference	± 4.15 ms
Communication and Overhead	± 3.01 ms
Total	± 65.67 ms

Table 4: Real-Time Performance

Table 4 shows that the system achieves an average end-to-end latency of 65.67 ms per inference cycle. This is a comparatively low figure for a system designed to operate in real time on 30 FPS webcam video, and it substantiates the central practical claim of this paper: that automatic apex-phase spotting and anxiety classification can be combined into a single pipeline fast enough for live, interactive use not merely for offline analysis.

5 Conclusion and Discussion

The results show that a TV-L1 optical flow based motion representation, combined with Region of Interest (ROI) constraints, is able to capture the low amplitude facial changes that characterize micro expressions. Restricting the analysis to the eyebrow, eye, and lip regions helps concentrate the analysis on the most informative areas of the face while simultaneously reducing computational complexity, which in turn supports real-time deployment.

On the phase detection front, the prominence based apex spotting method exhibits reasonably strong consistency on both the CASME II and SAMM datasets. Although the resulting MAE has not yet fully reached an ideal target, the evaluation shows that the majority of predictions land close to the true apex position. This finding indicates that the optical flow magnitude signal can serve as an effective indicator for automatically localizing the micro expression phase, an outcome that directly supports the feasibility of fully automatic, annotation free apex spotting as a building block for real-time systems.

At the classification stage, the Temporal CNN yields the highest validation accuracy, while the Transformer demonstrates stronger cross domain generalization. This contrast suggests that CNN based models are more effective at learning dataset specific patterns in the training data, whereas the Transformer is better able to sustain performance when confronted with variation across different subjects. Taken together with the latency results—an average end-to-end processing time of 65.67 ms—these findings substantiate the paper's central contribution: a complete, automatic, and genuinely real-time micro expression pipeline for anxiety recognition is achievable on ordinary web infrastructure, closing a gap that most prior offline, laboratory bound micro expression research has left open. Future work will focus on narrowing the validation to test generalization gap, most likely through larger and more diverse subject pools, domain adaptation techniques, and hybrid CNN-Transformer architectures, while preserving the real-time latency budget established in this study.

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